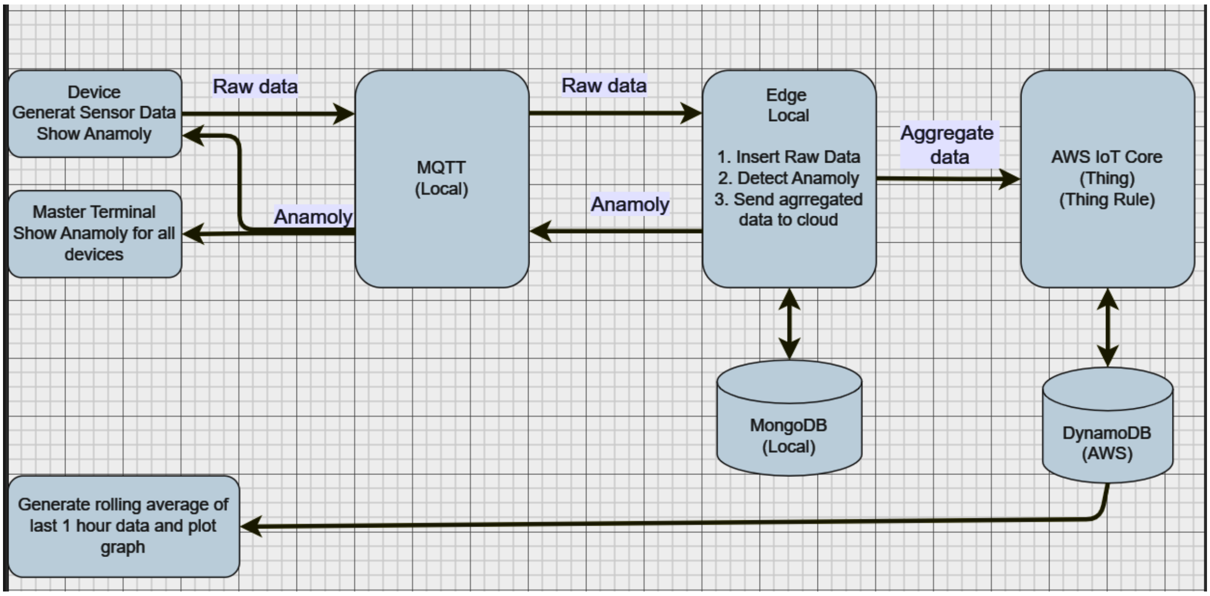
Project

The project focuses on integrating end to end components of IOT learned. We will implement a PoC of 4 tier architecture comprising of

1. Device
2. Edge
3. Cloud
4. Applications for Reporting/Visualization

Program Flow



1. Device (on local machine) – multiple devices
    a. Have at least 6 devices – 2 each to capture Heart rate, Blood Pressure and SPO2 levels.
    b. Generate Data of respective sensor type using simulator program.
        i. Each device generates 6 rows in a minute
        ii. There are total of 6 device – so 36 rows will be generated in a minute
    c. Each device will generate its own data (at interval of 10 seconds) and send to Edge via MQTT
    d. Display anomaly revived from Edge on respective device.
2. Edge (on local machine)
    a. Insert raw device data into Mongo DB table
    b. Detect anomaly – insert into another mongo DB table and also publish this back to respective device
    c. Aggregate and send aggregated data to cloud using AWS IoT Core
        i. Aggregation should be done every minute
        ii. There are 36 rows generated from 6 devices in 1 minute
        iii. This needs to be consolidated into total 6 rows i.e. 1 row per device with min, max and average for that device for that minute
3. Cloud (on AWS)
    a. Insert aggregated data received from Edge via AWS IoT Core into DynamoDB
4. Applications
    a. Master Terminal for monitoring by medical staff (on Local Machine) - Display

anomaly revived from Edge for all devices on a terminal.

b. Report & Visualization (local machine) - Pull out dynamo DB data for last hour or last day, create moving average and plot a graph on screen and save in html file.

Expectation from Learners

- We have done hands on for every components . Now learners will have utilize there learnings to build this system.
- Submit a video demonstrating working application and flow of code.